

Multiplying and Dividing Rational Expressions

As with numbers, rational expressions can be multiplied and divided.

Multiplying Rational Expressions

To multiply rational expressions, multiply the numerators and multiply the denominators. Divide out common factors to simplify.

EXAMPLE 1

Multiply.

A $\frac{3h}{(h-7)(h-2)} \cdot \frac{h-7}{h(h+8)}$

SOLUTION

$$\begin{aligned}\frac{3h}{(h-7)(h-2)} \cdot \frac{h-7}{h(h+8)} &= \frac{3h(h-7)}{(h-7)(h-2)(h)(h+8)} \\ &= \frac{3\cancel{h}(\cancel{h-7})}{(\cancel{h-7})(h-2)(\cancel{h})(h+8)} \\ &= \frac{3}{(h-2)(h+8)} \\ &= \frac{3}{h^2 + 6h - 16}\end{aligned}$$

The polynomials are in factored form, so multiply the numerators and the denominators.

Divide out common factors.

Simplify.

Alternate simplified form

► **Think About It** Both $\frac{3}{(h-2)(h+8)}$ and $\frac{3}{h^2 + 6h - 16}$ are in simplified form. Recognizing that they are equal can be helpful when you work with rational functions.

B $\frac{2x^2 - 10x}{x^2 - 2x - 8} \cdot \frac{x + 2}{6x^3 - 30x^2}$

SOLUTION

$$\begin{aligned} \frac{2x^2 - 10x}{x^2 - 2x - 8} \cdot \frac{x + 2}{6x^3 - 30x^2} &= \frac{2x(x - 5)}{(x + 2)(x - 4)} \cdot \frac{x + 2}{6x^2(x - 5)} \\ &= \frac{2x(x - 5)(x + 2)}{(x + 2)(x - 4)(6x^2)(x - 5)} \\ &= \frac{2\cancel{x}(x - 5)(x + 2)}{(\cancel{x + 2})(x - 4)(2\cancel{x} \cdot 3x)(x - 5)} \\ &= \frac{1}{(3x)(x - 4)} \\ &= \frac{1}{3x^2 - 12x} \end{aligned}$$

Factor the polynomials.

Multiply the numerators and the denominators.

Divide out common factors. To identify the greatest common monomial factor, write $6x^2$ as $2x \cdot 3x$.

Simplify.

Alternate simplified form ■

Dividing Rational Expressions

To divide by a rational expression, multiply by its reciprocal.

EXAMPLE 2

Divide.

A $\frac{5(x - 2)}{(x + 12)(x + 3)} \div \frac{15(x - 2)}{x(x + 12)}$

SOLUTION

Write the division problem as multiplying by a reciprocal.

$$\begin{aligned} \frac{5(x - 2)}{(x + 12)(x + 3)} \div \frac{15(x - 2)}{x(x + 12)} &= \frac{5(x - 2)}{(x + 12)(x + 3)} \cdot \frac{x(x + 12)}{15(x - 2)} \\ &= \frac{5\cancel{(x - 2)}(x)(x + 12)}{(\cancel{x + 12})(x + 3)(\cancel{15})(x - 2)} \\ &= \frac{x}{3(x + 3)} \text{ or } \frac{x}{3x + 9} \end{aligned}$$

Multiply by the reciprocal of the divisor.

Multiply numerators and denominators. Divide out common factors.

Simplify.

B $\frac{p^2 - 16}{4p^2 + 4p + 1} \div \frac{2p^2 - 7p - 4}{6p^2 + 3p}$

SOLUTION

Write the division problem as multiplying by a reciprocal.

$$\begin{aligned} \frac{p^2 - 16}{4p^2 + 4p + 1} \div \frac{2p^2 - 7p - 4}{6p^2 + 3p} &= \frac{p^2 - 16}{4p^2 + 4p + 1} \cdot \frac{6p^2 + 3p}{2p^2 - 7p - 4} \\ &= \frac{(p + 4)(p - 4)}{(2p + 1)(2p + 1)} \cdot \frac{3p(2p + 1)}{(p - 4)(2p + 1)} \\ &= \frac{(p + 4)(\cancel{p - 4})(3p)(\cancel{2p + 1})}{(2p + 1)(2p + 1)(\cancel{p - 4})(\cancel{2p + 1})} \\ &= \frac{3p(p + 4)}{(2p + 1)^2} \text{ or } \frac{3p^2 + 12p}{4p^2 + 4p + 1} \end{aligned}$$

Multiply by the reciprocal of the divisor.

Factor.

Multiply numerators and denominators.
Divide out common factors.

Simplify. ■